

# VINEET PRAKASH

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Streamlit Apps](#)

## PROFESSIONAL SUMMARY

IBM-certified Data Science professional with 7 production-deployed ML and GenAI projects spanning time-series forecasting, NLP, and RAG architectures. Built end-to-end pipelines from raw data to live Streamlit apps — including a RAG chatbot, churn predictor (AUC 0.829), and a recommendation system deployed on HuggingFace. Targeting demand forecasting and data science roles at product-first companies.

## SKILLS SUMMARY

- **Programming:** Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL, SQL Server)
- **Machine Learning / Deep learning / GenAI:** XGBoost, Scikit-learn, SMOTE, SHAP, Optuna, TF-IDF, SVD, TensorFlow, LangChain, RAG, FAISS, HuggingFace, Groq LLaMA 3, Vector DB, NLP, Prompt Engineering; **Time Series & Forecasting** Prophet, ARIMA, Lag Features, Rolling Averages, Seasonality Analysis.; **Data Analysis & EDA:** Feature Engineering, Leakage Detection, A/B Testing.
- **MLOps & Deployment:** Streamlit, Docker, Git & GitHub, ML Pipelines, CI/CD, FastAPI, HuggingFace Spaces.
- **Visualization & Tools:** Matplotlib, Seaborn, Power BI, Tableau, VS Code, Jupyter Notebook.

## EDUCATION

- **B.E. — Electrical, Electronics & Communications Engineering** (2018) Dayananda Sagar College of Engineering, Bangalore | Grade: A
- **IBM Data Science Professional Certificate** — Data Cleaning, ML, DL, AI (2026) [[credential link](#)]

## CERTIFICATIONS

- **Databases and SQL for Data Science with Python (IBM)** [[credential link](#)]
- **Programming for Everybody — University of Michigan:** Python, PostgreSQL, Web Scraping [[credential link](#)]
- **Google Prompting Essentials Specialization** — Prompt Engineering & LLM Design [[credential link](#)]
- **Career Essentials in Generative AI — Microsoft & LinkedIn:** Transformers, Fine-tuning, RAG Architectures [[credential link](#)]

## PROFESSIONAL EXPERIENCE

### British Airways — Forage [[Certificate](#)]

- Scraped & analyzed 1,000+ customer reviews using BeautifulSoup; built NLP pipeline — text cleaning, sentiment scoring & theme extraction from unstructured feedback.
- Developed classification model for customer buying behavior prediction; presented actionable insights to simulated leadership team with business recommendations.

### BCG GenAI Job Simulation—Forage [[Certificate](#)]

- Analyzed telecom client dataset to identify churn drivers using Python & Pandas; built Random Forest classifier framed as a business risk reduction problem.
- Delivered McKinsey-style executive presentation with data-driven recommendations — simulating end-to-end DS consulting engagement from problem framing to modeling.

## PROJECTS

### AskVineet — RAG Chatbot [[GitHub](#)] [[Live Demo](#)]

- Built a production-style RAG pipeline — PDF ingestion → chunk splitting → HuggingFace embeddings → FAISS similarity search → LLaMA 3 via Groq — with streaming responses, 3-turn chat history, multi-document querying & source page attribution, replicating core NotebookLM behaviour from scratch.
- Debugged LangChain v1beta API incompatibility with Gemini embeddings mid-build, identified root cause & pivoted to HuggingFace embeddings — eliminating rate limits, enabling offline inference, and improving production reliability.

### Recommendation System [[GitHub](#)] [[Live Demo](#)]

- Built a hybrid recommendation engine on 568K+ Amazon reviews — implemented SVD collaborative filtering, TF-IDF content-based & 60-40 weighted hybrid; solved 99.99% matrix sparsity via CSR Sparse Matrix, achieving Precision@10: 0.15, Recall@10: 0.30, NDCG@10: 0.32 (75x over random baseline).
- Added human-readable explainability per recommendation and hosted model on HuggingFace Datasets — delivering a production-style deployment with interpretable outputs, not just a notebook experiment.

### Sales Forecasting [[GitHub](#)] [[Live Demo](#)]

- Built a sales forecasting system on 3M+ rows across 54 stores & 33 product families — compared Prophet, XGBoost & ARIMA; Prophet achieved MAPE 9.98% (1.23% improvement over baseline), translating to ₹12L savings per ₹10Cr inventory.
- Engineered time-aware features (lag 7/14/28 days, rolling averages with look-ahead bias prevention, oil prices, holiday & promo flags); deployed Streamlit app with store-level forecasts, confidence intervals & What-If Simulator for promotion/holiday impact modeling.

### Customer Churn Prediction [[GitHub](#)] [[Live Demo](#)]

- Built end-to-end churn prediction on 7K+ IBM Telco customers using Logistic Regression, Random Forest & XGBoost with SMOTE; tuned via Optuna achieving AUC-ROC 0.829 & Recall 69.5% — flagging 573 high-risk customers with \$714K LTV and \$41.6K monthly revenue at stake.
- Engineered custom feature charge\_per\_service (top 5 SHAP driver); deployed Streamlit app with real-time churn probability, CLV score, priority tier & retention recommendations — making predictions directly actionable for non-technical business teams

### Loan Default Classifier [[GitHub](#)] [[Live Demo](#)]

- Built a loan default classifier on 200K LendingClub records — handled 80:20 class imbalance with SMOTE, engineered loan-to-income ratio & credit utilization features, and tuned XGBoost with Optuna achieving AUC-ROC 0.743.
- Applied SHAP explainability to surface top risk drivers per applicant for credit risk teams; deployed live Streamlit app enabling real-time default risk scoring for non-technical users without any code dependency.